

Proteostasis

Sérgio F genetic report
2 of October 2026 (local time)

Name	Sérgio F
Focus	longevity and immortality (84 aging mechanisms)
Data	Crude DNA (chip) + local imputation (only DR2 ≥ 0.8)
Content	17 Curated SNPs · 119 loci of GWAS · 5 polygenic scores · 5 321 pathogenic variants of ClinVar in 44 genes

What is: Sérgio F variants that affect the folding of proteins (chaperones, endoplasmic reticulum), their degradation (lysosomes, autophagy) and aggregation (amyloid, tau, alpha-synuclein, alpha-1 antitrypsin, transthyretin), with Parkinson's and Alzheimer's scores and a search for rare variants in 44 genes.

Privacy: the analysis was done locally; the genotypes never left the computer. Public databases were only sent SNP identifiers.

Important: genetic predispositions, not a diagnosis. The conclusions come only from the genes (without analysis or exams). All practical ideas are suggestion (not applied), to decide with the doctor.

Contents

1. Abstract: favourable and unfavourable	2
2. Method, chain and trust.....	2
3. Polygenic scores	2
4. Favourable	3
5. Unfavourable	3
6. Rare variants (ClinVar) and chip boundaries.....	5
7. GWAS loci	6
8. Map for 84 aging mechanisms.....	8
9. Practical section — suggestion (not applied)	9
10. Limitations.....	9
11. Full table SNP to SNP	10

1. Abstract: favourable and unfavourable

Favourable	Unfavourable
• APOE ε3/ε3: without ε4, the greatest genetic accelerator of amyloid aggregation. • MAPT H2/H2 (rs8070723 GG): tau protective haplotype (less supranuclear palsy and Parkinson). • Non carrier of Parkinson's lysosomal variants (GBA1 E326K, N370S, T369M; LRRK2 G2019S) or the amyloidogenic transthyretin V122I. • PERK (EIF2AK3) rs7571971 CC: without the risk allele of response to reticulum stress. • Parkinson: 39 loci score at the 28th percentile (and 35 in the 90 variants); SNCA and MAPT favourable.	• SERPINA1 Pi*MZ probable (rs28929474 CT): alpha-1 antitrypsin Z folds poorly and accumulates in the liver reticulum; ~35% lower levels. • Alzheimer without APOE (39 variants) at the 91st percentile; the 69 loci score and genomics are typical (47 and 52 percentiles). • 16 Alzheimer's loci and unfavourable Parkinson's 11 (e.g. ABCA7, MS4A4A, TMEM175, LRRK2 rs76904798, GPNMB). • TREM2 R47H SNP not found in the data.

Reading for immortality: Neuronal proteostasis is globally favourable (APOE ε3/ε3, MAPT H2/H2, without Parkinson's lysosomal variants), but there is a clear weakness in the liver and lung: alpha-1 antitrypsin Z. For immortality, the priority is to protect the liver (which already has high ALT and PNPLA3) and the lung, and maintain active cleaning pathways (autophagy, sleep, exercise).

2. Method, chain and trust

- **Genotypes:** 96 read directly on the chip (GRCh38, strand +); 29 locally imputed (Beagle 5.5, full genome reference panel 30x, only DR2 ≥ 0.8); 11 without data ("SNP not found in the data").
- **Strand:** All alleles were aligned to the GRCh38 strand + and verified against alleles and frequencies of the reference panel. 1 alleles of the literature were in the strand coding – and were converted (complement); 10 palindromic SNPs (A/T or C/G) were resolved by the allelic frequency; the non-resolvable palindromic SNPs were out of the scores.
- **Classification:** compares the copies of the effect allele with the average of the reference population (2 × frequency). Difference ≥ 0.5 copies → Favourable or Unfavourable; lower → Typical. Not having a rare variant of risk, or a common variant of strong effect, counts as Favourable. An allele with opposite effects (e.g. longer telomeres but higher risk of cancer) appears as Commitment in both lists.
- **Trust:** chip > imputation (DR2 ≥ 0.8; estimated accuracy of ~95–99% in common variants, lower in rare ones). Evidence: Strong (replicated in large/functional GWAS), Moderate, Weak (small or inconsistent studies).
- **Polygenic scores:** weighted sum of alleles (published coefficients); percentile calculated against the reference population with exactly the same variants.
- **Sources of interpretation:** dbSNP/Ensembl, ClinVar, GWAS Catalog, PGS Catalog and articles cited in the text.

3. Polygenic scores

Score	Percentile	Variants used	Interpretation	Note
Parkinson's disease (39 loci, Chang 2017) G_PD · calculated in this report	28	39 of 39 (100%)	Favourable	Low percentile : less polygenic risk.
Parkinson's disease (90 variants, Nalls 2019) PGS000902 · calculated in this report	35	87 of 90 (97%)	Typical	Typical, in the same sense.
Alzheimer's disease (69 loci, Bellengez 2022) G_AD · calculated in this report	47	69 of 72 (96%)	Typical	Typical.
Alzheimer without APOE (39 variants) PGS000811 · previous genomic imputation (same method)	91	37 of 39 (95%)	Unfavourable	High percentile; the genomic score (249 thousand variants) is typical.
Alzheimer's (genomic, 249 thousand variants) PGS002249 · previous genomic imputation (same method)	52	212 109 of 249 273 (85%)	Typical	Typical.

the 50th percentile : median of the reference population. For risk scores, high percentile = higher risk; for "pro" scores (e.g. longer telomeres, longevity), high percentile = more of the trait. Typical = percentile between 30 and 70. The scores explain only a small part of the variation and are not diagnostic.

4. Favourable

Gene/score	SNP	Genotype	Source · evidence	Effect
APOE	rs429358 ε4 (Cys112Arg) allele	TT 0× C	imputation (DR2=0.99) Strong	Without the C-risk allele. The ε4 accelerates the aggregation of beta-amyloid, worsens the lysosomal clearance and the blood-brain barrier; it is the largest common genetic factor of Alzheimer. Mechanisms: Protein aggregation; Neuroinflammation; Hematoencephalic Barrier dysfunction; Lysosome dysfunction
MAPT	rs8070723 haplotype H1/H2	GG 0× A	chip Moderate	Without the risk A allele. The A allele marks the H1 haplotype, which increases tau expression and the risk of progressive supranuclear paralysis and Parkinson's. H2 is the protective haplotype. Mechanisms: Protein Aggregation
GBA1	rs2230288 Glu326Lys (E326K)	CC 0× T	chip Moderate	Without the T-risk allele. Variant of lysosomal glucocerebrosidase that reduces enzymatic activity and increases the risk of Parkinson's disease by about 1.6–2. Mechanisms: Lysosome dysfunction; Protein Aggregation
GBA1	rs76763715 Asn370Ser (N370S)	TT 0× C	chip Strong	Without the C-risk allele. Gaucher variant; in heterozygosity, the risk of Parkinson's disease increases about 3–5. Mechanisms: Lysosome dysfunction; Protein Aggregation
GBA1	rs75548401 Thr369Met (T369M)	GG 0× A	chip Weak	Without the risk A allele. Mild variation in glucocerebrosidase; slightly increased risk of Parkinson's disease. Mechanisms: Lysosome dysfunction
LRRK2	rs34637584 Gly2019Ser (G2019S)	GG 0× A	chip Strong	Without the risk A allele. Variant of gain in function of LRRK2 kinase that disturbs lysosomes and autophagy; main cause of genetic Parkinson. Mechanisms: Lysosome dysfunction; Autophagy / Deregulated Macroautophagy; Protein Aggregation
TTR	rs76992529 Val142Ile (V122I)	GG 0× A	chip Strong	Without the risk A allele. Amyloidogenic variant of transthyretin (cardiac amyloidosis). Mechanisms: Protein Aggregation
EIF2AK3	rs7571971 PERK (UPR response)	CC 0× T	chip Moderate	Without the T-risk allele. The T allele alters PERK, the response arm to poorly folded proteins in the endoplasmic reticulum; it is associated with progressive supranuclear paralysis (Höglinger 2011). Mechanisms: Endoplasmic Reticulum Disorder; Remodelling of the Endoplasmic Reticulum
polygenic	G_PD Parkinson's disease (39 loci, Chang 2017)	the 28th percentile	39 variants Score	Low percentile : less polygenic risk. Mechanisms: Protein aggregation; Lysosome dysfunction
SNCA, MAPT, HMG2P18, MCCC1, INPP5F, RIT2	Parkinson's disease 11 of 40 loci		GWAS GWAS	11 In which the genotype of Sérgio F deviates from the mean of the reference population in the favourable sense (favourable = less Parkinson's risk). Mechanisms: Protein aggregation; Lysosome dysfunction; Neuroinflammation
SLC24A4, USP6NL-AS1, INPP5D, LINC02498, ABI3, NCK2	Alzheimer's disease 17 of 79 loci		GWAS GWAS	17 In which the genotype of Sérgio F deviates from the mean of the reference population in the favourable sense (favourable = less risk of Alzheimer). Mechanisms: Protein aggregation; Neuroinflammation; Lysosome dysfunction

5. Unfavourable

Gene/score	SNP	Genotype	Source · evidence	Effect
SERPINA1	rs28929474 Glu342Lys (Pi*Z allele)	TC 1× T	chip Strong	1 copy of risk allele T. The Z allele folds poorly and polymerizes in the endoplasmic reticulum of the hepatocyte; in carriers (Pi*MZ) the levels of alpha-1 antitrypsin decrease about 35% and the risk of liver fibrosis rises, especially with fatty liver. Mechanisms: Protein aggregation; Endoplasmic Reticulum Disorder; Remodelling of the Endoplasmic Reticulum ER-fagia Defectosa; Fragmentation of Elastin; Proteostasis loss
polygenic	PGS000811 Alzheimer without APOE (39 variants)	the 91st percentile	37 variants Score	High percentile; the genomic score (249 thousand variants) is typical. Mechanisms: Protein aggregation; Neuroinflammation

Gene/score	SNP	Genotype	Source · evidence	Effect
TMEM175, NUCKS1, LRRK2, BST1, GPNMB, NDUFAF2	Parkinson's disease 11 of 40 loci		GWAS GWAS	11 In which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = less Parkinson's risk). Mechanisms: Protein aggregation; Lysosome dysfunction; Neuroinflammation
MS4A4A, ABCA7, PMS2P1, CD2AP, MINDY2-DT, EPHA1-AS1	Alzheimer's disease 16 of 79 loci		GWAS GWAS	16 In which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = less risk of Alzheimer). Mechanisms: Protein aggregation; Neuroinflammation; Lysosome dysfunction

6. Rare variants (ClinVar) and chip boundaries

All 5 321 variants classified as pathogenic or probably pathogenic in ClinVar were searched for in the 44 genes of the panel of this report (ABCA7, APP, ATP13A2, BAG3, C9orf72, CHCHD10, CRYAB, CTSD, DNAJB6, FUS, GAA, GBA1, GLA, GRN, HEXA, HSPB1, HSPD8, LAMP2, LRRK2, MAPT, NPC1, NPC2, OPTN, PARK7, PINK1, PRKN, PRNP, PSEN1, PSEN2, PSMB8, SEPINA1, SNCA, SOD1, SORL1, SQSTM1, TARDBP, TBK1, TREM2, TTR, UBA1, UBQLN2, VCP, VPS35). Of these, 538 have probes on the chip and additional 10 are in the imputation panel.

Gene	Variant	Genotype	Condition (ClinVar)	Evaluation
GBA1	rs439898 1:155238630 G>A	AA	GBA1-related disorder; not provided; Gaucher disease; Gaucher disease type II	Homozygosity would be Gaucher's disease; the region has the pseudogene GBAP1, which confuses the probes. Artifact.
SERPINA1	rs28929474 14:94378610 C>T	CT	SERPINA1-related disorder; Susceptibility to severe coronavirus disease (COVID-19); Neurodevelopmental disorder; Inborn gen	Probable Pi*MZ (Z allele carrier): plausible and already known. Confirm by alpha-1 serum antitrypsin + Pi phenotyping.

Chip limit: the chips genotype well common variants, but often fail in the rare ones (false positives and false negatives; Weedon 2021). The absence of pathogenic variants here does not exclude hereditary syndromes. Only clinical sequencing (panel, exome or genome) in a certified laboratory does so.

7. GWAS loci

For each study: independent loci (one SNP per region of 500 kb, the most significant, $p \leq 5 \times 10^{-8}$), with the genotype of Sérgio F and the classification against the reference population. The typical loci are in the complete table (CSV); here we show those that move away from the average.

Parkinson's disease

Chang D et al. 2017 (20 184 cases). Effect allele: more Parkinson's risk = unfavourable. 40 loci: 11 favourable, 11 unfavourable, 18 typical, 0 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
SNCA	rs356182	G ↑	AA	0.35	chip	Favourable
MAPT	rs17649553	C ↑	TT	0.76	imputation (DR2=1.00)	Favourable
TMEM175	rs34311866	C ↑	CT	0.19	chip	Unfavourable
HMG2P18 - KRTCAP2	rs35749011	A ↑	GG	0.02	chip	Favourable
MCCC1	rs12637471	G ↑	GA	0.80	chip	Favourable
NUCKS1 - RAB29	rs823118	T ↑	TT	0.55	chip	Unfavourable
LRRK2, LRRK2-DT	rs76904798	T ↑	TT	0.13	chip	Unfavourable
BST1	rs11724635	A ↑	AA	0.55	chip	Unfavourable
INPP5F	rs117896735	A ↑	GG	0.01	imputation (DR2=1.00)	Favourable
GPNMB	rs199347	A ↑	AA	0.62	chip	Unfavourable
RIT2	rs12456492	G ↑	AA	0.34	chip	Favourable
NDUFAF2	rs2694528	C ↑	CA	0.11	chip	Unfavourable
SH3GL2	rs13294100	G ↑	GG	0.66	chip	Unfavourable
CTSB	rs2740594	A ↑	AA	0.73	imputation (DR2=0.99)	Unfavourable
MSR1 - RN7SL474P	rs591323	G ↑	GG	0.71	imputation (DR2=0.99)	Unfavourable
ITPKB	rs4653767	T ↑	TT	0.68	chip	Unfavourable
MAP4K4	rs34043159	C ↑	TT	0.35	chip	Favourable
SYT17	rs11343	T ↑	GG	0.47	chip	Favourable
DLG2	rs3793947	G ↑	AA	0.56	chip	Favourable
IP6K2	rs12497850	T ↑	TT	0.65	chip	Unfavourable
ITIH1	rs143918452	A ↑	GG	0.00	chip	Favourable
BIN3	rs2280104	T ↑	CC	0.37	chip	Favourable

Alzheimer's disease

Bellenguez C et al. 2022 (111 326 cases). Effect allele: more risk of Alzheimer : unfavourable. 79 loci: 17 favourable, 16 unfavourable, 36 typical, 10 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
MS4A4A	rs1582763	A ↓	GG	0.38	imputation (DR2=0.99)	Unfavourable
ABCA7	rs12151021	A ↑	AA	0.34	imputation (DR2=0.99)	Unfavourable
PMS2P1	rs7384878	C ↓	TT	0.32	chip	Unfavourable
CD2AP	rs7767350	T ↑	TC	0.25	imputation (DR2=0.99)	Unfavourable
SLC24A4	rs12590654	A ↓	AA	0.33	chip	Favourable
USP6NL-AS1 - ECHDC3	rs7912495	G ↑	AA	0.45	chip	Favourable
INPP5D	rs10933431	G ↓	GC	0.23	chip	Favourable
LINC02498 - MIR572	rs6846529	C ↑	TT	0.29	chip	Favourable
MINDY2-DT	rs602602	A ↓	TT	0.31	chip	Unfavourable
ABI3	rs616338	T ↑	CC	0.01	chip	Favourable
EPHA1-AS1	rs11771145	A ↓	GG	0.36	chip	Unfavourable
PLCG2	rs12446759	G ↓	AA	0.39	chip	Unfavourable
NCK2	rs143080277	C ↑	TT	0.01	chip	Favourable
DOC2A	rs1140239	T ↓	CC	0.40	imputation (DR2=0.99)	Unfavourable
ZNF594-DT, SCIMP	rs7225151	A ↑	AG	0.14	chip	Unfavourable
WNT3	rs199515	G ↓	GG	0.22	chip	Favourable
APP	rs2154481	C ↓	CC	0.49	imputation (DR2=0.98)	Favourable

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
IDUA, SLC26A1	rs3822030	G ↓	TT	0.43	chip	Unfavourable
BCKDK	rs889555	T ↓	CC	0.29	chip	Unfavourable
MME	rs61762319	G ↑	AA	0.02	chip	Favourable
RNU6-1307P - LILRB2	rs587709	C ↑	TT	0.32	imputation (DR2=0.97)	Favourable
TMEM106B	rs13237518	A ↓	AA	0.40	chip	Favourable
UMAD1	rs6943429	T ↑	CC	0.42	imputation (DR2=0.98)	Favourable
CYYR1 - ADAMTS1	rs2830489	T ↓	CC	0.29	imputation (DR2=0.95)	Unfavourable
SEC61G-DT - EGFR	rs76928645	T ↓	TC	0.13	chip	Favourable
GPR141 - NME8	rs6966331	T ↓	CC	0.36	chip	Unfavourable
IL34	rs4985556	A ↑	AC	0.11	imputation (DR2=0.94)	Unfavourable
HDAC2-AS2, HS3ST5	rs785129	T ↑	CC	0.33	chip	Favourable
SLC2A4RG	rs6742	T ↓	TC	0.21	chip	Favourable
PRKD3	rs17020490	C ↑	CT	0.17	imputation (DR2=1.00)	Unfavourable
SORT1	rs141749679	C ↑	TT	0.00	chip	Favourable
TNIP1	rs871269	T ↓	CC	0.35	chip	Unfavourable
LINC00917 - FENDRR	rs16941239	A ↑	TT	0.03	chip	Favourable

8. Map for 84 aging mechanisms

Mechanism	Favourable	Unfavourable	Interpretation
Protein Aggregation	APOE rs429358; MAPT rs8070723; GBA1 rs2230288; GBA1 rs76763715; LRRK2 rs34637584; TTR rs76992529; G_PD; Parkinson's disease; Alzheimer's disease	SERPINA1 rs28929474; PGS000811; Parkinson's disease; Alzheimer's disease	Favourable neuron (APOE, MAPT, GBA1/LRRK2); unfavourable liver (Pi*Z).
Neuroinflammation	APOE rs429358; Parkinson's disease; Alzheimer's disease	PGS000811; Parkinson's disease; Alzheimer's disease	APOE ε3/ε3 favourable; Alzheimer's without unfavourable APOE.
Hematoencephalic Barrier dysfunction	APOE rs429358	—	APOE ε3/ε3 (without ε4).
Lysosome dysfunction	APOE rs429358; GBA1 rs2230288; GBA1 rs76763715; GBA1 rs75548401; LRRK2 rs34637584; G_PD; Parkinson's disease; Alzheimer's disease	Parkinson's disease; Alzheimer's disease	GBA1/LRRK2 non-carrier (favourable); TMEM106B/GRN on average.
Autophagy / Deregulated Macroautophagy	LRRK2 rs34637584	—	Predominant
Endoplasmic Reticulum Disorder	EIF2AK3 rs7571971	SERPINA1 rs28929474	Pi*Z: poorly folded protein in the hepatocyte reticulum (unfavourable); PERK favourable.
Remodelling of the Endoplasmic Reticulum	EIF2AK3 rs7571971	SERPINA1 rs28929474	Pi*Z (unfavourable).
ER-fagia Defective	—	SERPINA1 rs28929474	Pi*Z: depends on ER-phagy to clean polymers (unfavourable).
Fragmentation of Elastin	—	SERPINA1 rs28929474	Pi*Z: less inhibition of elastase (unfavourable).
Proteostasis loss	—	SERPINA1 rs28929474	Pi*Z.
Molecular Platerone Failure	—	—	HSP70 (HSPA1L) informative; HSPA1B without data.

9. Practical section — suggestion (not applied)

Coherent with the existing genetic plan and the current stack reported. Nothing was applied here; everything is to discuss with the doctor.

#	Suggestion	Why (genes)	State
1	Confirm Pi*MZ: alpha-1 serum antitrypsin + Pi phenotyping	Single plausible variant; changes the surveillance of the liver and lung.	suggestion (not applied)
2	Protecting the liver: maintaining curcumin, luteolin, apigenin and quercetin pauses while ALT is high (plan)	The Pi*Z hepatocyte already accumulates poorly folded protein.	suggestion (not applied)
3	Protecting the lung: no smoking, vaping or dust without mask	Less alpha-1 antitrypsin : less protection of lung elastin.	suggestion (not applied)
4	Sleep of 7-9 h	Glymphatic cleaning of amyloid and tau occurs mainly in deep sleep.	suggestion (not applied)
5	Regular exercise and heat (sauna) if tolerated	They induce heat shock chaperones (HSP70) and autophagy.	suggestion (not applied)
6	Maintain lithium 5 mg and rapamycin (already in the stack)	Lithium: less tau phosphorylation (GSK3) and more autophagy; rapamycin: autophagy.	suggestion (not applied)
7	Omega-3 to 3.2-4 g/day with DHA (already taking fish oil) and more fiber	DHA in neuronal membranes; less inflammation.	suggestion (not applied)
8	Maintain L-carnosine (already in the stack)	It reduces glycation and aggregation of proteins.	suggestion (not applied)

Avoiding or with caution

- Hepatotoxic supplements (green tea extract/EGCG, kava, red rice yeast), as in the plan. *(suggestion, not applied)*
- Repeated cranial trauma (unprotected contact sports). *(suggestion, not applied)*

10. Limitations

- TREM2 R47H is not in the data; rare variants of APP/PSEN and repetitions (C9orf72, HTT) are not testable on the chip.
- Pi*MZ is probable but not clinically confirmed.
- The GWAS loci show average associations in the population; the "mapped" gene is the closest and not always the causal gene.
- The effects of isolated SNPs are small; several studies with only one population. They do not allow clinical conclusions by themselves.
- Imputation: an imputed genotype may be wrong, especially in rare variants; the imputed ones are marked.
- This report uses only genes: analysis, exams or measurements were not used.

11. Full table SNP to SNP (cured SNPs)

Gene	SNP	Effect allele (+)	Genotype	Freq.	Source	Chain	Class	Evidence
SERPINA1	rs28929474 Glu342Lys (Pi*Z allele)	T (risk)	TC	0.017	chip	strand + (without conversion)	Unfavourable	Strong
SERPINA1	rs17580 Glu264Val (Pi*S allele)	A (risk)	TT	0.060	chip	palindromic: confirmed by frequency (reference population 0.06)	Typical	Moderate
APOE	rs429358 ε4 (Cys112Arg) allele	C (risk)	TT	0.153	imputation (DR2=0.99)	strand + (without conversion)	Favourable	Strong
APOE	rs7412 ε2 (Arg158Cys) allele	T (protector)	CC	0.066	chip	strand + (without conversion)	Typical	Strong
MAPT	rs8070723 haplotype H1/H2	A (risk)	GG	0.763	chip	strand + (without conversion)	Favourable	Moderate
GBA1	rs2230288 Glu326Lys (E326K)	T (risk)	CC	0.018	chip	strand + (without conversion)	Favourable	Moderate
GBA1	rs76763715 Asn370Ser (N370S)	C (risk)	TT	0.002	chip	strand + (without conversion)	Favourable	Strong
GBA1	rs75548401 Thr369Met (T369M)	A (risk)	GG	0.006	chip	strand + (without conversion)	Favourable	Weak
LRRK2	rs34637584 Gly2019Ser (G2019S)	A (risk)	GG	0.000	chip	strand + (without conversion)	Favourable	Strong
TREM2	rs75932628 Arg47His (R47H)	T (risk)	—	0.006	SNP not found in the data	strand + (without conversion)	Undetermined	Strong
TTR	rs76992529 Val142Ile (V122I)	A (risk)	GG	0.000	chip	strand + (without conversion)	Favourable	Strong
EIF2AK3	rs7571971 PERK (UPR response)	T (risk)	CC	0.298	chip	strand + (without conversion)	Favourable	Moderate
GRN	rs5848 3'UTR (progranulin)	T (risk)	TC	0.296	chip	strand + (without conversion)	Typical	Weak
TMEM106B	rs1990622 TMEM106B (lysosomes)	A (risk)	AG	0.597	imputation (DR2=0.99)	strand + (without conversion)	Typical	Moderate
PRNP	rs1799990 Met129Val	A (informative)	AG	0.673	chip	strand + (without conversion)	Information	Moderate
HSPA1L	rs2227956 Met493Thr (HSP70-hom)	G (informative)	GA	0.160	chip	Allele of the literature in strand – (coding); converted to strand +	Information	Weak
HSPA1B	rs1061581 +1267 A>G (HSP70-2)	G (informative)	—	0.400	SNP not found in the data	strand + (without conversion)	Undetermined	Weak

Notes about the typical, informative and without data: SERPINA1 rs17580 (TT): The S allele folds poorly (less than Z); combined with Z (Pi*SZ) gives relevant deficit. APOE rs7412 (CC): The ε2 is associated with less amyloid and more longevity. TREM2 rs75932628 (SNP not found in the data): It reduces the function of microglia in the removal of amyloid; increases 2–3 times the risk of Alzheimer's disease. GRN rs5848 (TC): The T allele is associated with less progranulin (lysosomal protein) and, in some studies, frontotemporal degeneration. TMEM106B rs1990622 (AG): Allele A increases TMEM106B (lysosomal protein) and the risk of frontotemporal degeneration with TDP-43; G is the protective allele. PRNP rs1799990 (AG): Met129 in homozygosity increases the susceptibility to prion diseases (very rare); heterozygosity is protective. HSPA1L rs2227956 (GA): Chaperone of the HSP70 family; small associations and no established direction. HSPA1B rs1061581 (SNP not found in the data): Synonymous variant of HSP70-2; inconsistent associations.